AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 6 · STUDENT LESSON

# Sound Becomes a Signal

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

## Keep the values and their times.

Trace a sound record, compare equal maxima, inspect omitted samples and construct a new hidden change.

---

**Keep one habit:** Name the sample times → trace the retained information → limit the interpretation.

---

MAIN LESSON · SECTION 1

## What do the samples keep about a sound?

> Keep each recorded value with its time before interpreting a pattern.

### Start with equal maxima

Two lists contain the same sound-signal values, but their largest values occur at different times. A maximum tells you how high a recorded value reached; it does not tell you when it happened.

## Retrieve the image-count limit

Two image grids have the same number of black pixels. Why can their pictures differ?

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

## By the end, you can...

- I can interpret an ordered sequence of sound-**signal sample**s with a stated time interval.
- I can explain why summaries or sparse **sample**s can omit changes.
- I can separate an audio recording, a **transcript** and an interpretation of meaning.

### ▼ Earlier wording, alternatives and picture transcripts

#### Your learning target

#### Remember before reading

Without opening the old lesson, explain one key idea from Pictures Become Numbers. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Sensors Are Messengers. Give an example and a mistake that the idea helps you avoid.

Computers receive changing measurements, not human-like hearing.



LEARNING TARGET A whisper recorded close to a microphone has larger **sample amplitude**s than a shout recorded far away. Does the recording prove the whisper was louder at its source? Propose one controlled follow-up.

Learner name:

### Readiness for the connected explanation

### This week’s end goal

By the end, I can connect ordered sound **sample**s to timing and explain what a maximum loses.

MAIN LESSON · SECTION 2

# A microphone records a changing signal

> Our tiny invented sequences let you inspect information loss on paper.

## From pressure variation to ordered measurements

Sound involves changing pressure in a medium such as air. A microphone responds to that variation and produces a changing **signal**. A digital recording stores measurements of the **signal** at successive times.

The values are ordered: swapping their times changes the represented pattern.

### Read the toy-value convention

We will use tiny invented sequences to study this idea on paper. Their values are arbitrary **signal** units, not decibels or a test of safe listening levels.

Real audio uses many more **sample**s, so our five-point examples should not be played as if they were a realistic recording of speech.

## Keep the timing metadata

A recording must include enough information to interpret the values, including their time spacing and encoding. The sequence 0,2,0,2,0 does not by itself say how quickly the changes happened.

This is similar to needing image dimensions as well as pixel values.

## Check the record

- Which information tells you the order of the samples? Which tells you how quickly they were taken? Explain what is missing if someone sends only 0,2,0,2,0.

## ▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

### Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



MYSTERY A microphone converts air-pressure changes into a sequence of measured values that can be plotted as a **waveform**.

### What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

### A closer explanation

# Recording, transcription and action are different stages

> Trace the intermediate record instead of assuming a microphone understands words.

## Follow the information path

### Three stages in a speech system

| INPUT | OPERATION | OUTPUT |
| --- | --- | --- |
| Sound reaching a microphone | Measure the changing signal at successive times | Ordered samples and timing information |
| Recorded audio | Speech recognition estimates the words | A transcript that may contain errors |
| Transcript and an explicit request | A separate rule or model processes the text | An answer or proposed action |

Recording, transcribing and interpreting are separate operations. A later stage can correctly process an incorrectly transcribed number.

## Count the gaps between time labels

Time in milliseconds · 0 · 1 · 2 · 3 · 4

> **Time in milliseconds:** Invented signal value
>
> **0:** 0
>
> **1:** 2
>
> **2:** 0

> **3:** 2
>
> **4:** 0

There are five **sample**s with four gaps between the first and last. Each gap is 1 millisecond, so the elapsed time from first to last is 4 milliseconds.

If the same five values were spaced 2 milliseconds apart, they would span 8 milliseconds. Equal value lists with different timing do not describe the same time pattern.

## Explain an arrow

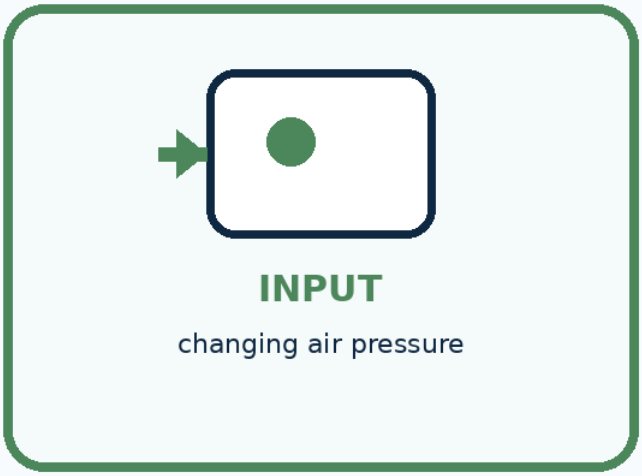
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

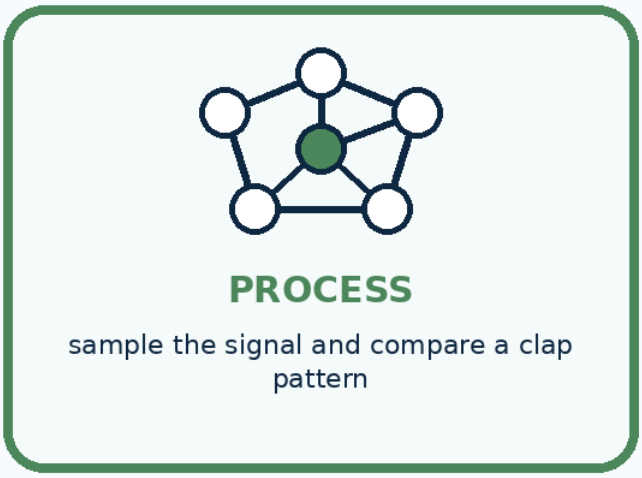
### ▼ Earlier wording, alternatives and picture transcripts

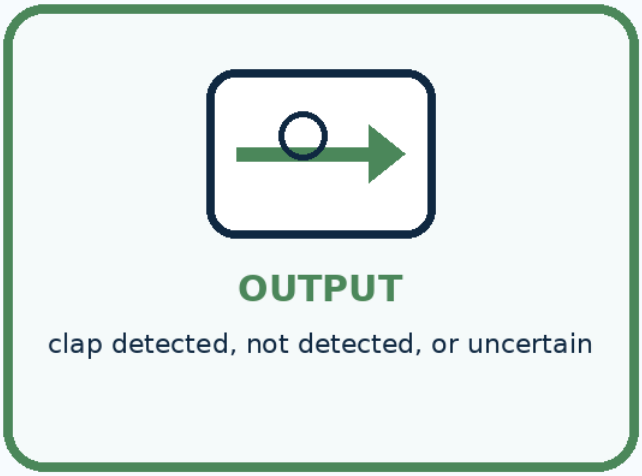
VISUAL **MECHANISM**

**Input → Process → Output**

Follow the information instead of imagining magic inside the machine.






**INPUT** changing air pressure

**PROCESS sample** the **signal** and compare a clap pattern

**OUTPUT** clap detected, not detected, or uncertain

CORE IDEA Computers receive changing measurements, not human-like hearing.

UNDER THE HOOD Audio is represented by **sample**s taken over time. **Sample** rate describes how often measurements are taken; **amplitude** describes **signal** strength.

**Noise** can overlap the target pattern, so detection requires a decision rule or model and an uncertainty policy.

Draw a fresh example with all three stages:

**A closer explanation**

**Connect the picture and the explanation**

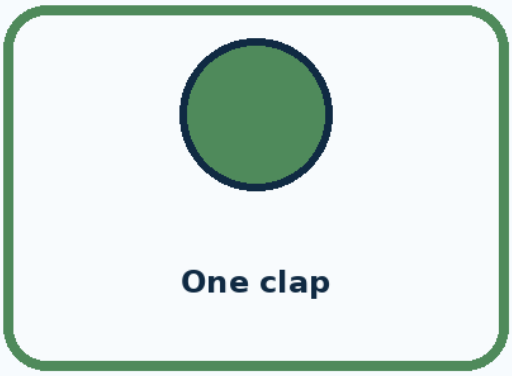
---

MAIN LESSON · SECTION 4

## Compare the recorded patterns

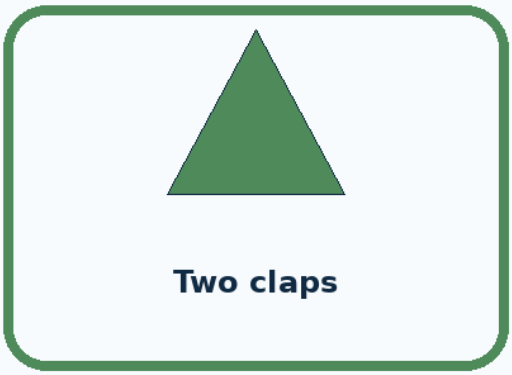
> Use the timing and shape clues without claiming that one peak identifies a sound’s meaning.

## Pictures paired with their evidence




**The case to investigate**

CASE 1 — One clap Clue: single sharp peak. Meaning: target pattern.




**The case to investigate**

CASE 2 — Two claps Clue: two separated peaks. Meaning: different timing pattern.




**The case to investigate**

CASE 3 — Talking Clue: many irregular changes. Meaning: possible false trigger.




**The case to investigate**

CASE 4 — Quiet room Clue: small background values. Meaning: baseline.

## Discuss the confusing pair

- Which two pictures are easiest to confuse? What **evidence** separates them?

### Keep the limit in view

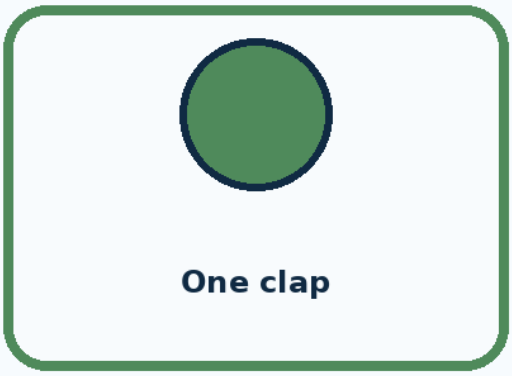
BOUNDARY A microphone measures a **signal**; it does not understand a speaker's meaning by itself.

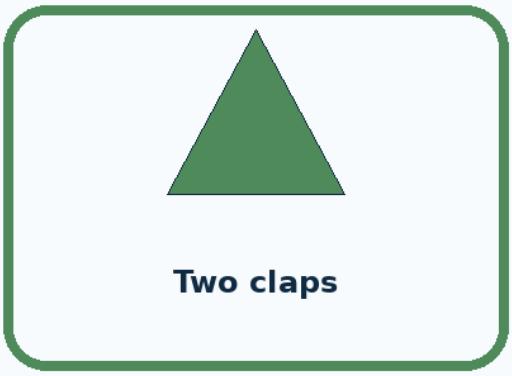
▼ Earlier wording, alternatives and picture transcripts

PICTURE GALLERY

#### Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









CASE 1 — One clap Clue: single sharp peak. Meaning: target pattern.

CASE 2 — Two claps Clue: two separated peaks. Meaning: different timing pattern.

CASE 3 — Talking Clue: many irregular changes. Meaning: possible false trigger.

CASE 4 — Quiet room Clue: small background values. Meaning: baseline.

### Change one thing in the pictured case

Change: Retain only **sample**s at 0,2,4 ms.

**Prediction**: Both retained sequences are 0,0,0.

Reason: The selected times omit both nonzero **sample**s from each **signal**.

# Same maximum. Different recorded timing.

> Keep the time axis and distinguish a full sequence from a summary.

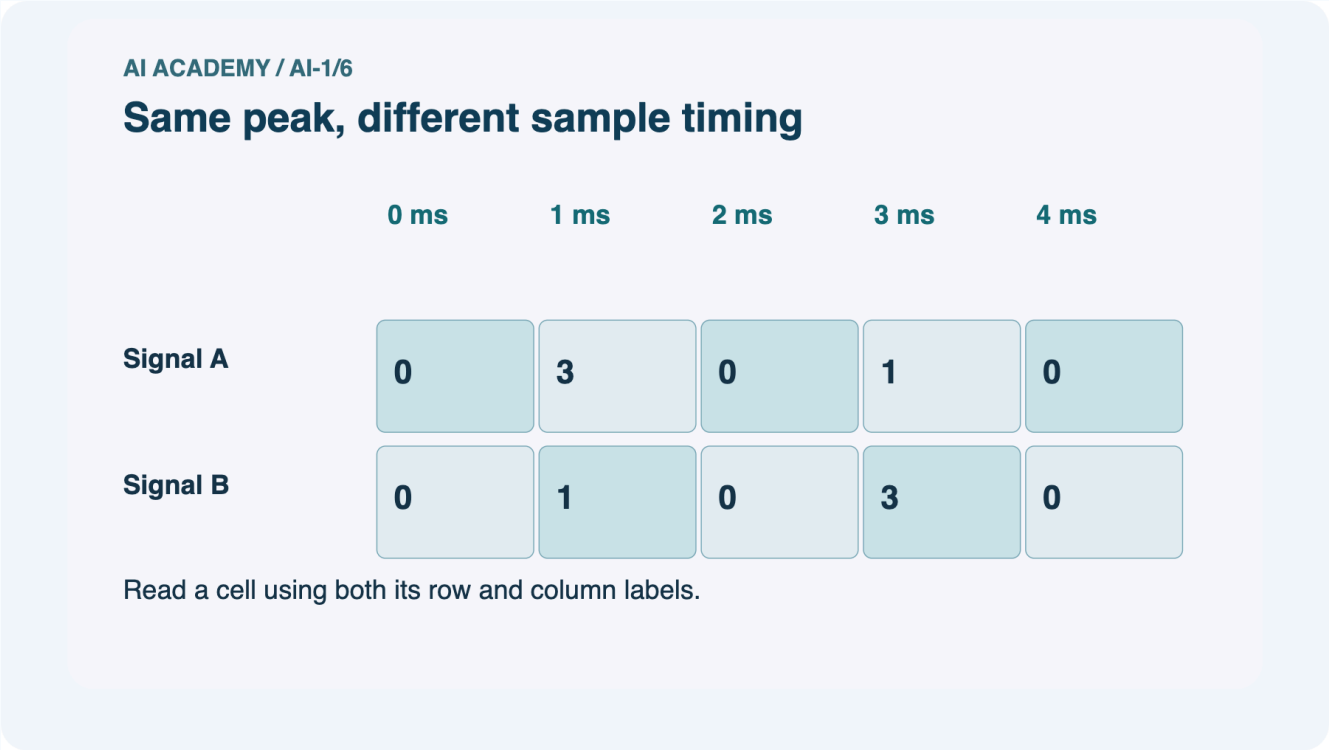
## Given times and toy signal values

A fictional microphone display records five **sample**s during a short classroom tap.

Times are 0,1,2,3,4 ms in order.

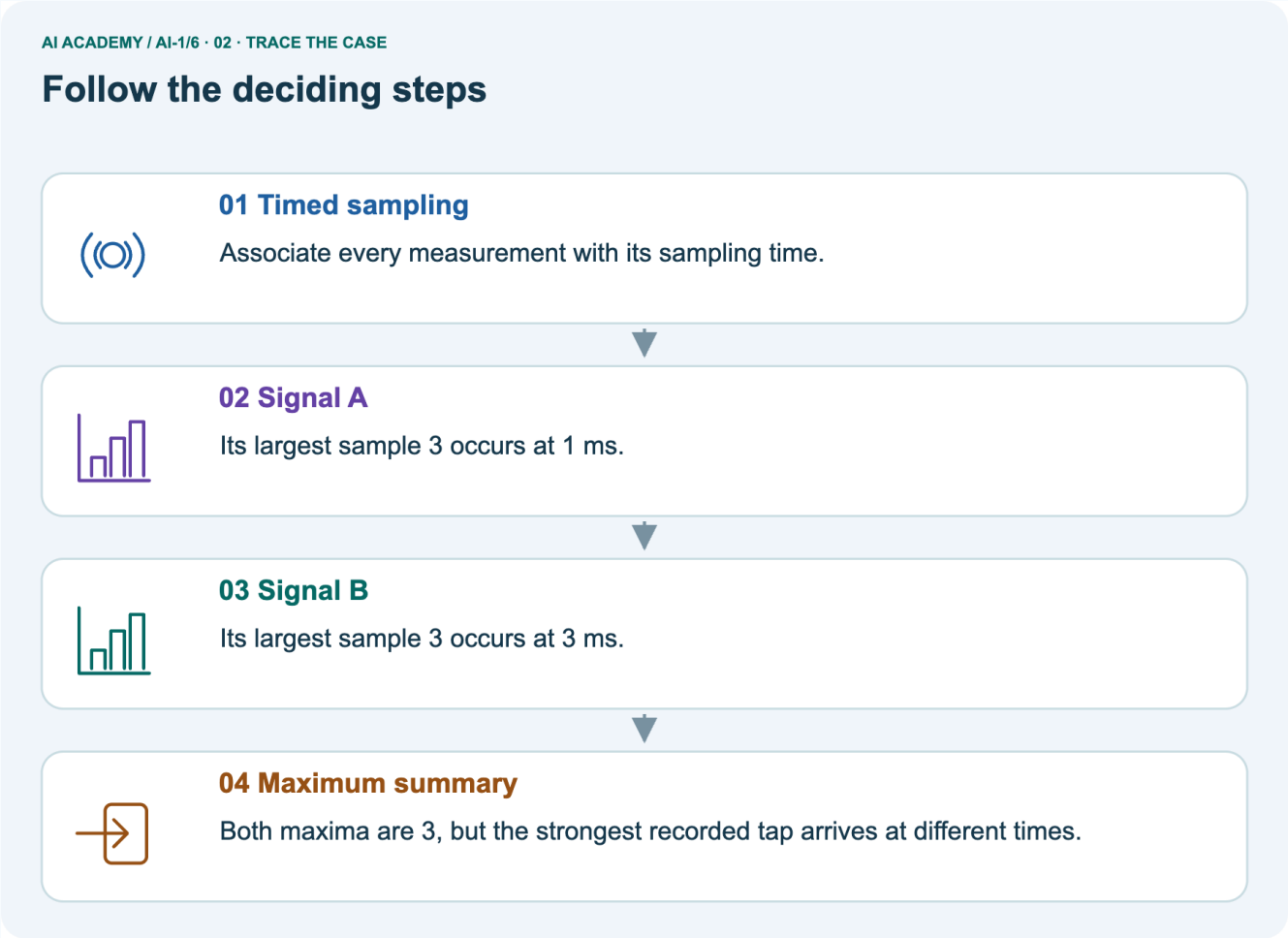
**Signal** A: 0,3,0,1,0. **Signal** B: 0,1,0,3,0. Toy values are signed measurement units; these happen to be nonnegative.

## Compare where the recorded peaks occur




Follow the labelled picture with the matching explanation. Open the vector diagram

## Trace the maximum summary




Follow the labelled picture with the matching explanation. Open the vector diagram

## Explain what the maximum loses

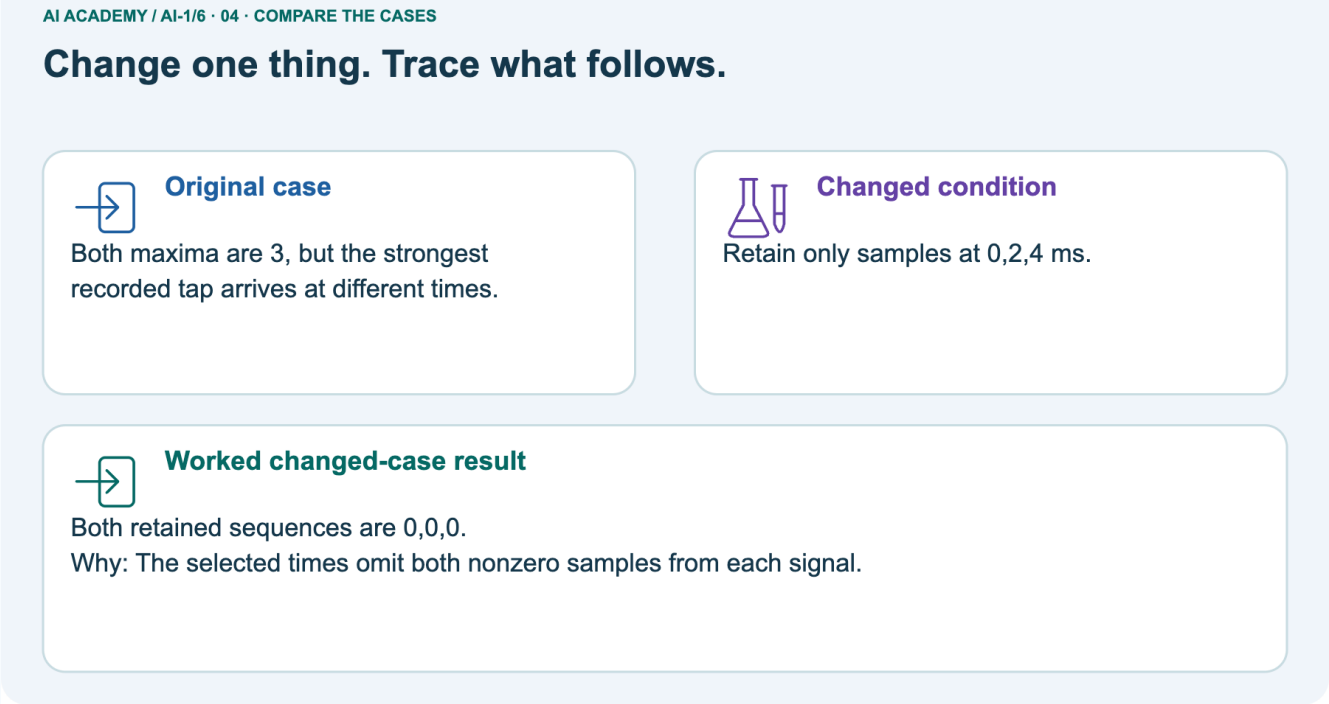
The sequences contain the same values in different orders. Keeping the maximum preserves peak recorded value while losing its timing.

A graph should keep time on the horizontal axis; shuffling **sample**s changes the recorded **signal** even if its maximum stays fixed.

## Predict the sparse record

- Before checking, keep only times 0,2,4 ms in each supplied signal. List each retained sequence. Can those values alone establish what happened at 1 and 3 ms?

## ▼ Check the retained samples




Follow the labelled picture with the matching explanation. Open the vector diagram

### Limit the claim about real sound

These sparse toy values cannot establish perceived loudness, pitch or what happened between sampling times.

## Explain the omitted changes

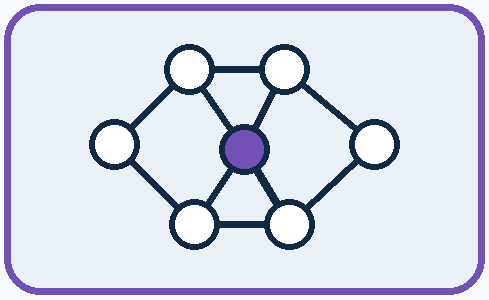
- Read the labelled picture: Does the all-zero retained record prove the full **signal** was silent? Use the picture.

## ▼ Earlier wording, alternatives and picture transcripts

WORKED EXAMPLE

### Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



CASE A microphone converts air-pressure changes into a sequence of measured values that can be plotted as a **waveform**.

STEP 1 — OBSERVE changing air pressure

STEP 2 — TRACE **sample** the **signal** and compare a clap pattern

STEP 3 — CONCLUDE clap detected, not detected, or uncertain

MODEL REASONING Look for two strong peaks separated by the expected gap. **Noise** may add extra peaks or hide a clap. A safe detector needs thresholds, timing rules, and a not-sure result when **evidence** is weak.

DO NOT OVERCLAIM A microphone measures a **signal**; it does not understand a speaker's meaning by itself.

What new **evidence** could change this answer?

### A closer explanation

The largest value in 0,2,0,2,0 is 2. The largest value in 0,0,2,0,0 is also 2. The first sequence reaches 2 twice at different times, while the second does so once.

Keeping only the maximum loses that distinction, just as counting black pixels lost their arrangement.

Sparse measurements can also miss changes. If we inspect the first sequence only at times 0, 2 and 4 milliseconds, all retained values are 0. The two intervening values of 2 disappear from the record.

It would be incorrect to conclude from these retained **sample**s alone that the **signal** stayed at zero between them.

More frequent sampling can preserve faster changes under suitable recording conditions, but sampling frequency alone does not make every recording clear.

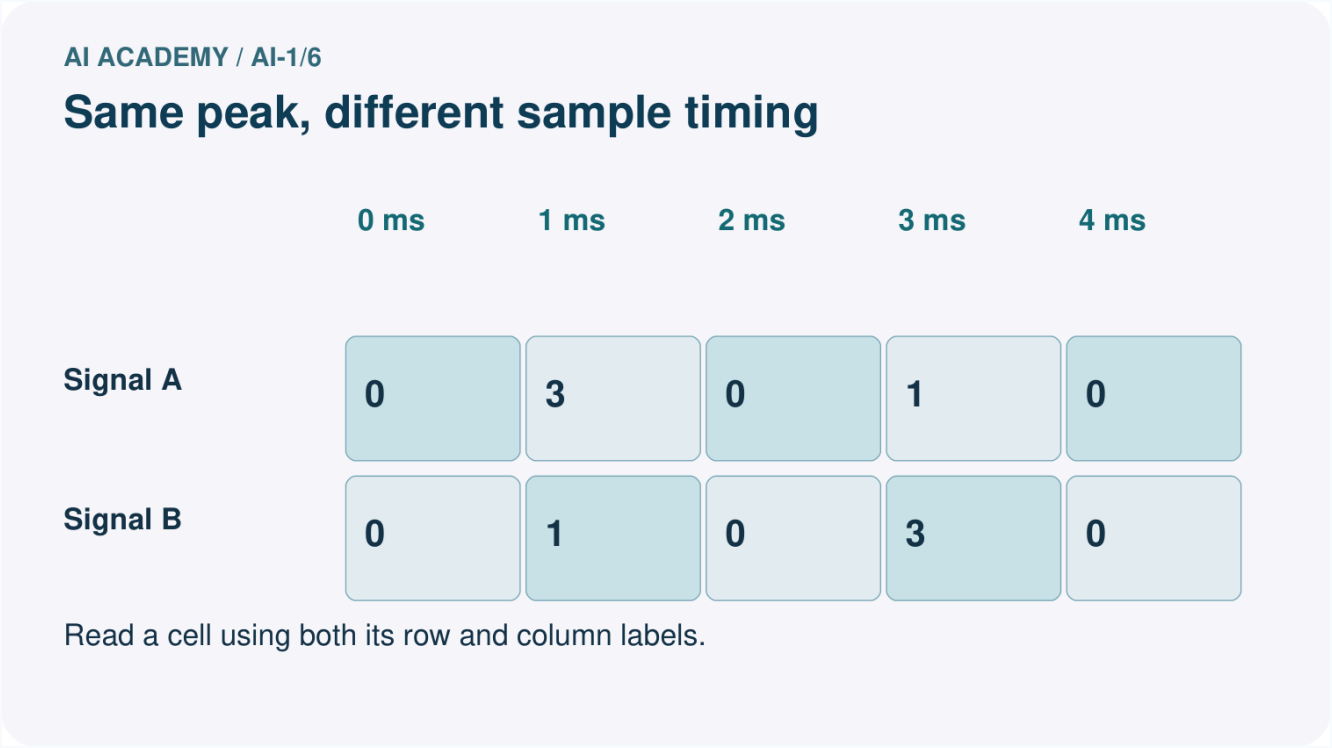
Microphone placement, background sound, clipping and the recording format can also matter. We are studying one source of missing information, not a complete audio engineering specification.

### Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

### A pictorial worked case: Sound Becomes a Signal

### The facts we will use




Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/6

# Sound Becomes a Signal

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



## 1. Timed sampling

Associate every measurement with its sampling time.



## 2. Signal A

Its largest sample 3 occurs at 1 ms.



## 3. Signal B

Its largest sample 3 occurs at 3 ms.



## 4. Maximum summary

Both maxima are 3, but the strongest recorded tap arrives at different times.

**CHANGE ONE THING**

Retain only samples at 0,2,4 ms.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Timed sampling | Associate every measurement with its sampling time. |
| 2. Signal A | Its largest sample 3 occurs at 1 ms. |
| 3. Signal B | Its largest sample 3 occurs at 3 ms. |
| 4. Maximum summary | Both maxima are 3, but the strongest recorded tap arrives at different times. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

### What this example does not establish

Change one thing: Retain only **sample**s at 0,2,4 ms. Predict the result before checking the explanation. What remains the same?

---

MAIN LESSON · SECTION 7

## Find a transcription error before acting

> A correct calculation can still use the wrong interpreted number.

## A transcript is a different representation

A **transcript** might record the words Fine, thank you while omitting a pause, a laugh or overlapping speech. It might also record the wrong words.

Text can be useful for searching spoken content, but it is not the same representation as the audio. Do not infer a speaker's emotions with certainty from a short **transcript**.

### Trace sixteen becoming sixty

In a voice calculator, audio saying sixteen can be transcribed as sixty. A calculator can then correctly divide sixty by four and return fifteen. Inspecting the intermediate **transcript** identifies the first mismatch.

Confirming the interpreted numbers before acting can prevent this particular error from travelling into a later stage.

## Repair and explain

- CLAIM A A loud sound must be a clap.
- CLAIM B Silence means the sensor has no data.

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

**Explain one detail a transcript can preserve and one sound detail it may omit. Explain why a correct calculation does not prove the transcript was correct**

▼ **Earlier wording, alternatives and picture transcripts**

COMPARE

### Same Result, Different Reason

A comparison helps reveal the true concept boundary.

| Case | Evidence | What it shows |
|---|---|---|
| One clap | single sharp peak | target pattern |
| Two claps | two separated peaks | different timing pattern |
| Talking | many irregular changes | possible false trigger |
| Quiet room | small background values | baseline |

### Repair two tempting claims

Repair:

Repair:

### A closer explanation

---

MAIN LESSON · SECTION 8

## Practise: count samples, gaps and retained values

> These practice samples use a 2 ms interval; read that new timing before calculating.

### The practice record and your trace

A five-**sample** record has values 0,3,0,1,0 at times 0,2,4,6,8 milliseconds. Read the timing before answering. The first-to-last duration is the final time minus the initial time, not the **sample** count multiplied by the gap.

> **State the interval and first-to-last duration. If only times 0,4,8 are kept, list the retained values and identify a change the shorter record misses**

## Construct another possible hidden pattern

To test a claim about the spaces between **sample**s, construct another possible sequence that agrees at retained times but differs between them.

If both match the stored record, that record alone cannot decide which intervening pattern occurred.

## Use W2 and explain to a partner

Use workbook W2 for the connected example “Sound as **sample**d information”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

| Plan | Do | Check |
|---|---|---|
| What is given? Which rule or procedure applies? | Record each intermediate step. | Does the result follow? What remains unknown? |

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

## Challenge “five samples make five gaps”

- Use the worked case for Sound as **sample**d information. Proposed explanation: “Five **sample**s make five gaps”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

## Check the pictured omission

Does the all-zero retained record prove the full **signal** was silent? Use the picture.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

GUIDED PRACTICE

### Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: Computers receive changing measurements, not human-like hearing.

Use a fresh example to explain **waveform**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: Silence means the sensor has no data.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

**A closer explanation**

**Practise with less help: Inspect what was discarded**

**Check your reasoning**

**Explain the picture**

---

MAIN LESSON · SECTION 9

# Investigate a fully specified paper pattern detector

> Predict, trace and record the rule's result, including a plateau and missing data.

### Your detector and its stated limits

Use five paper sample cards at 0,100,200,300,400 ms. Define an interior peak as a value at least 3 with both neighbouring values strictly smaller. Our toy rule says MATCH only when there are exactly two interior peaks 200 ms apart; otherwise NO MATCH. If any value is missing, say NOT SURE. MATCH means this numeric pattern matched, not that two real claps were proved.

## Run the three supplied cases

1. Predict before applying the rule: A=0,4,0,4,0; B=0,4,4,4,0; C=0,4,?,4,0. Mark the peak positions and compare the time gap. Let a partner apply the rule and record the actual paper trace. Discuss how a background sound could mimic the same values. No recording or loud sound is needed.

2. Keep a normal case and an edge case clearly separated.
3. Record what happened—even when it does not match your **prediction**.
4. Explain the result, one limitation, and the next useful test.

## Keep predictions, traces and uncertainty

| Case | Prediction | Observed result | What it means |
|---|---|---|---|
| Normal | | | |
| Edge | | | |

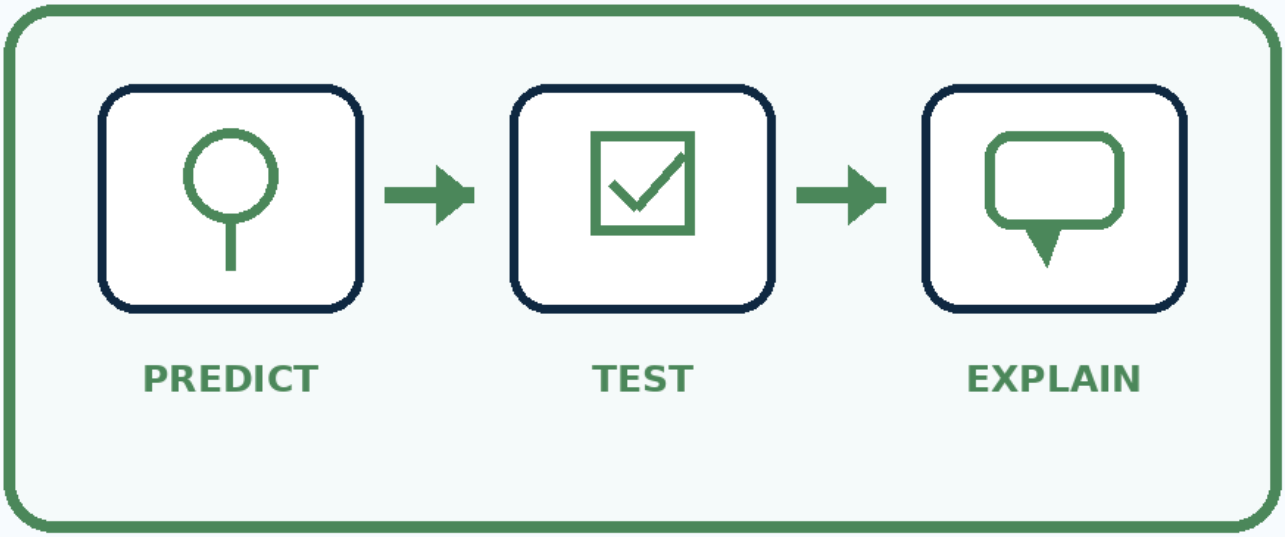
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

### ▼ Earlier wording, alternatives and picture transcripts

EXPLORER LAB

#### Clap Pattern Detector

Predict first, test honestly, and explain what the **evidence** means.




State your **prediction** before the result is known.

Complete the task: Build a clap-pattern detector; test it with background **noise**.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

**Plan → try → check**

---

MAIN LESSON · SECTION 14

## Try alone: construct a new hidden change

> Use every whole-millisecond position from 0 to 6 and preserve all three retained zeros.

### The new case

A record keeps values 0 at times 0,3,6 milliseconds. Construct two seven-**sample** sequences for every whole-millisecond time from 0 through 6 that agree with those retained values but differ at omitted times.

Explain why the retained record cannot prove that nothing changed between measurements.

**My independent explanation, trace and limits**

## Keep one first attempt

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

## Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

### Independent case: Construct a new hidden change

### Week 6 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Two recordings contain the same word: one quiet and one loud. Which device receives the sound? Does a larger wave automatically mean a different word? Explain.

My answer, with a trace, calculation or supporting **evidence**

**Write or draw your reasoning and evidence.**

### Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

**Write or draw your reasoning and evidence.**

### Connect to your project

If the sorter uses sound, specify the exact sound pattern needed and test whether classroom **noise** makes that **input** unreliable.

The **evidence** I will keep

**Write or draw your reasoning and evidence.**

### Transfer practice from the original programme

A whisper recorded close to a microphone has larger **sample amplitude**s than a shout recorded far away. Does the recording prove the whisper was louder at its source? Propose one controlled follow-up.

> **Explain the mechanism, evidence and one limit.**

**Use feedback to improve**

---

MAIN LESSON · SECTION 11

# Keep the timing your sorter needs

> Use sound only when a defined pattern serves the classroom task.

## This week’s project decision

THIS WEEK'S CHECKPOINT If the sorter uses sound, specify the exact sound pattern needed and test whether classroom **noise** makes that **input** unreliable.

Record the sorter’s **sampling interval** and retain relevant sequences when a decision depends on changes over time.

## Plan information, tests and review

- What real classroom need does this serve?
- What information is necessary—and what should stay private?
- What happens inside the **process**?
- What normal and edge cases will you test?

- When should it say not sure or ask a person?

## Keep measurement separate from meaning

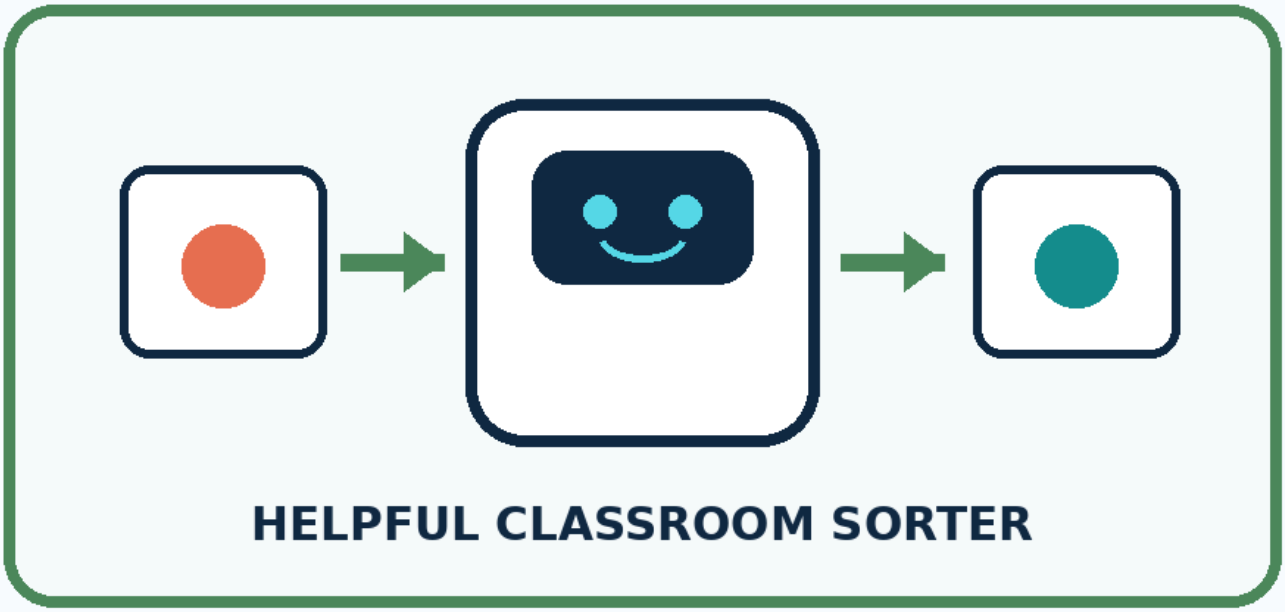
RESPONSIBLE DESIGN A microphone measures a **signal**; it does not understand a speaker's meaning by itself.

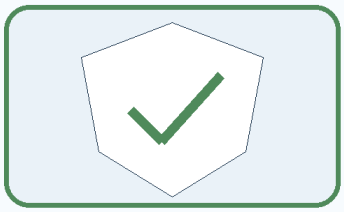
▼ **Earlier wording, alternatives and picture transcripts**

PROJECT

### Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.



My project decision:

**Evidence for your project**

---

MAIN LESSON · SECTION 16

# Finish: explain timing and information loss

> Close your notes and distinguish the stored record from a claim about the original sound.

## Your exit explanation

- Close your notes. Explain this goal in your own words: Interpret an ordered sequence of sound-**signal sample**s with a stated time interval

> **My deciding step, evidence and one remaining question**
>
> | |
> |---|
> | |
> | |
> | |
> | |

## Choose support and return later

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

# Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

## Your quick word guide

> Use the words to explain a particular step, record or limit.

### Core vocabulary

| Word | Meaning | My example |
|---|---|---|
| waveform | graph of a signal changing over time | |
| sample | one measurement at one moment | |
| amplitude | size or strength of a signal value | |
| noise | unwanted signal mixed with the target | |

### Terms for the connected explanation

| Word | Meaning |
|---|---|
| Signal | A changing physical or recorded quantity that can carry information |
| Sampling interval | The time between successive recorded samples |
| Transcript | A text representation of spoken words, which may contain errors or omit sound details |

## Repair a claim using a word

- CLAIM TO REPAIR A loud sound must be a clap.
- Use one vocabulary word and one piece of **evidence** to repair it:

▼ **Earlier wording, alternatives and picture transcripts**

VOCABULARY

### Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.












**Misconception detective**

WORD POWER A strong definition helps you decide something in a new case.

**Words used in the closer explanation**

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

## Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

## Pick your next step

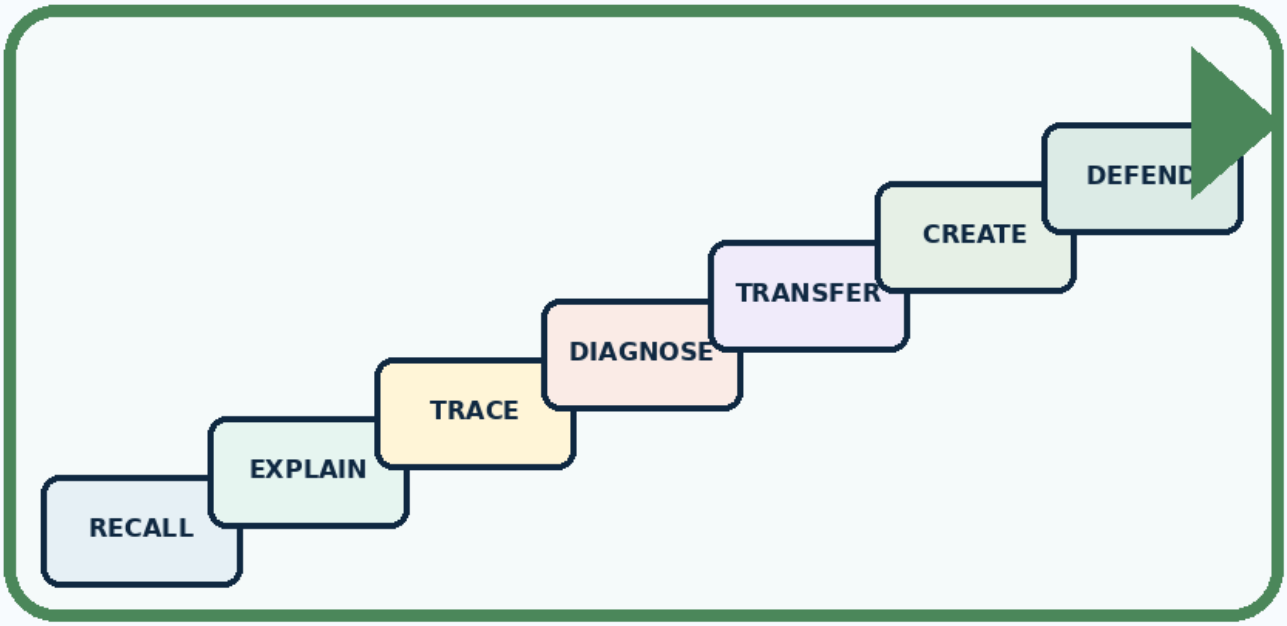
- RECALL Define **waveform**.
- EXPLAIN Computers receive changing measurements, not human-like hearing.
- TRACE Show the **input**, **process**, and **output**.
- DIAGNOSE Repair: A loud sound must be a clap.
- TRANSFER Apply the idea to a new home or classroom case.
- CREATE Design one revealing test.
- DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

### ▼ Earlier wording, alternatives and picture transcripts

PRACTICE LADDER

#### Climb from Recall to Defense

Each step asks for a stronger kind of understanding.




RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

▼ **Reference: check your explanation**

REFERENCE · SECTION 12

# Reference: check your explanation

> Use after practice. Construct a new hidden change is the selected independent case.

## Check the evidence and its limit

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.

- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

## Record support

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

### ▼ Earlier wording, alternatives and picture transcripts

MASTERY

#### Checklist for explaining this week's idea

Use after practice. Construct a new hidden change is the selected independent case.




EXPLANATION CHECK Keep sample values with their times; calculate gaps and duration, trace any discarded samples and state what remains unknown.

Independent **evidence**:

SELF-CHECK Look for two strong peaks separated by the expected gap. **Noise** may add extra peaks or hide a clap. A safe detector needs thresholds, timing rules, and a not-sure result when **evidence** is weak.

NEXT BRIDGE Carry today's **evidence** habit into: Categories and Labels.

▼ Extra practice: Sound samples show changes in amplitude

REFERENCE · SECTION 13

# Extra practice: Sound samples show changes in amplitude

> Try the question before opening its worked explanation.

### Read the supplied case and question

A microphone records quiet and loud claps at the same distance and settings.

What **waveform** difference is expected? Does it reveal a spoken word?

## Your explanation

- Which detail supports the answer, and why does it matter?

**Write or draw your reasoning and evidence.**

## ▼ Check after trying

The louder clap generally produces larger **amplitude**. **Amplitude** is not word meaning. Distance, gain and **noise** must be controlled.

| Evidence or stage | Value or result |
| --- | --- |
| Quiet clap | 1 |
| Loud clap | 3 |

## ▼ Earlier wording, alternatives and picture transcripts

### Week 6 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

### The situation

### The question

### Worked explanation

### Explain it in your own words

### Trace the supplied case visually

### Sound samples show changes in amplitude

A microphone records quiet and loud claps at the same distance and settings.

What **waveform** difference is expected? Does it reveal a spoken word?

The louder clap generally produces larger **amplitude**. **Amplitude** is not word meaning. Distance, gain and **noise** must be controlled.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

# Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

## Targets and evidence

| By the end, I can | How I will show it |
|---|---|
| Interpret an ordered sequence of sound-signal samples with a stated time interval | Explain the deciding step, show a trace or test, and state one limit. |
| Explain why summaries or sparse samples can omit changes | Explain the deciding step, show a trace or test, and state one limit. |
| Separate an audio recording, a transcript and an interpretation of meaning | Explain the deciding step, show a trace or test, and state one limit. |

## Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional microphone display records five **sample**s during a short classroom tap.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

# Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

## First supported question

- A fictional microphone display records five **sample**s during a short classroom tap.
- Which labelled step matches this detail? Both maxima are 3, but the strongest recorded tap arrives at different times.
- 1. **Signal** A
- 2. **Signal** B
- 3. Maximum summary
- 4. Timed sampling
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Practice 1: explain the deciding evidence

**My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Maximum summary Both maxima are 3, but the strongest recorded tap arrives at different times. The sequences contain the same values in different orders.

Keeping the maximum preserves peak recorded value while losing its timing. A graph should keep time on the horizontal axis; shuffling **sample**s changes the recorded **signal** even if its maximum stays fixed.

## Changed-case question

- Changed-case practice: Retain only **sample**s at 0,2,4 ms. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. Both retained sequences are 0,0,0.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed case**

After your attempt, compare with the supported explanation: Both retained sequences are 0,0,0. The selected times omit both nonzero **sample**s from each **signal**.

## Repair and save your work

Review Pictures Become Numbers (ai-1/5). Goal: Encode and reconstruct a small image using an explicit grid convention.

Short practice: Find a position where the original badges differ, then read the changed A back as a picture.. Return to this week and explain what you changed.

Review Sensors Are Messengers (ai-1/3). Goal: Describe a measurement using its quantity, value, unit and conditions. Short practice: Why can we calculate the earlier gap but not predict an exact new B reading?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

**Repair a missing building block**

**Keep your completed work**

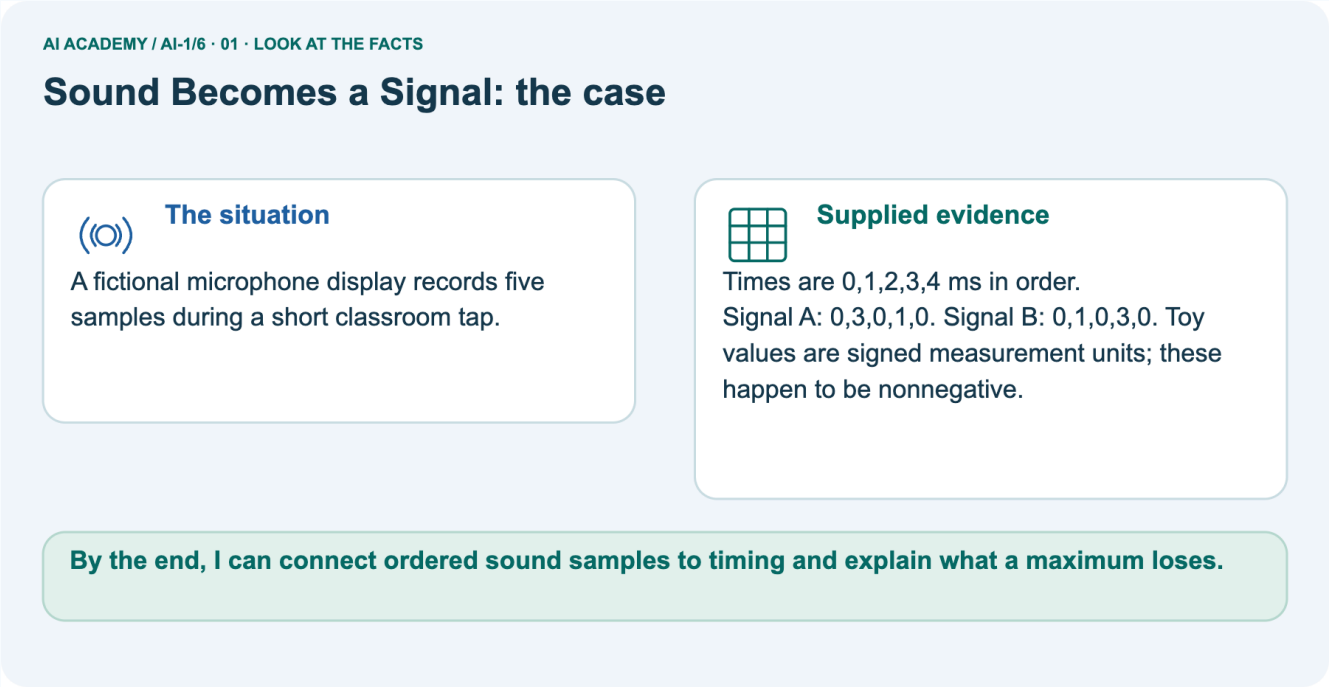
▼ **Reference: the six-picture journey**

REFERENCE · SECTION 18

# Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

## 01. Sound Becomes a Signal: the case



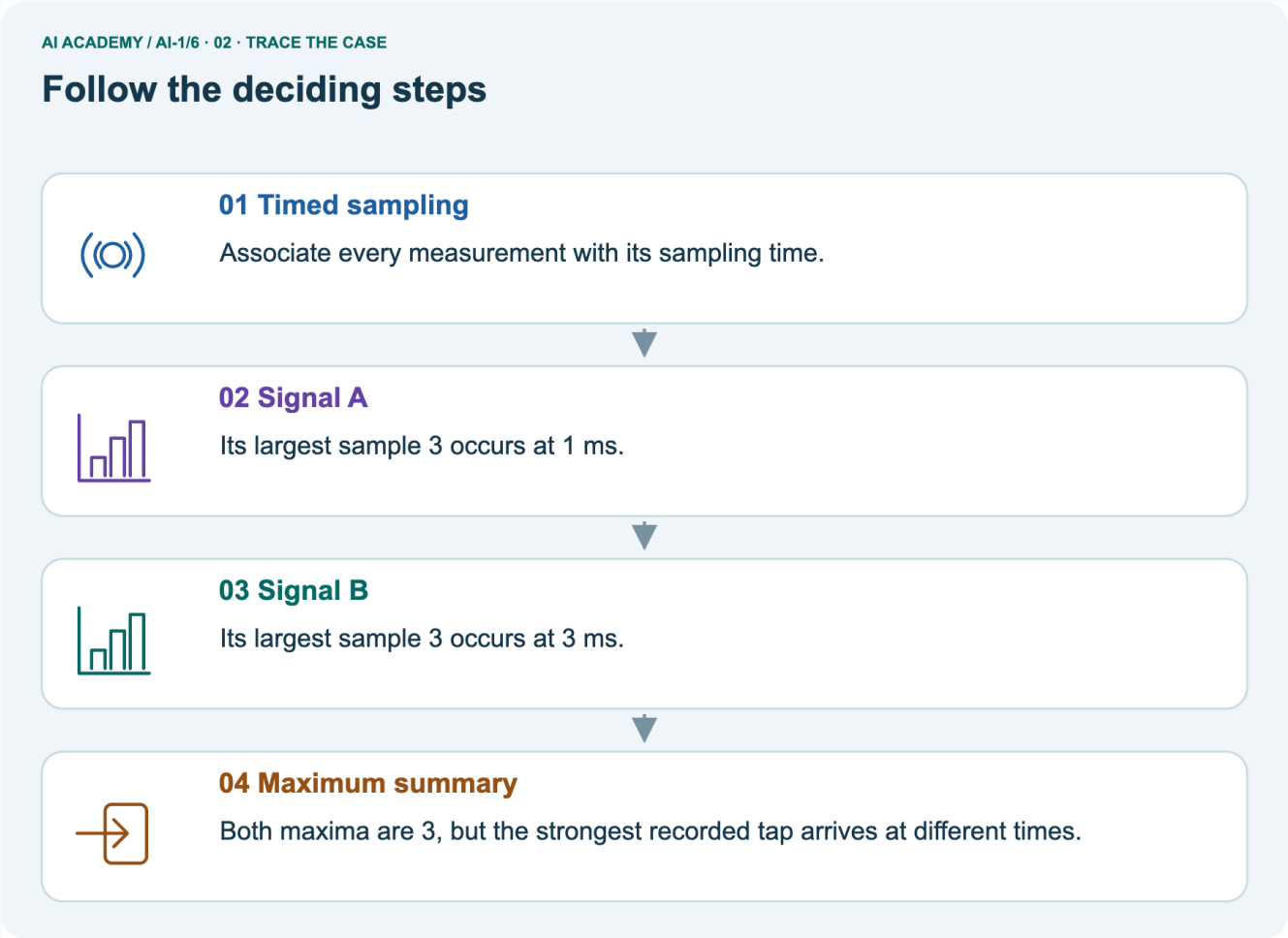

Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional microphone display records five **sample**s during a short classroom tap. Supplied facts: Times are 0,1,2,3,4 ms in order. **Signal** A: 0,3,0,1,0. **Signal** B: 0,1,0,3,0.

Toy values are signed measurement units; these happen to be nonnegative.

Point to the given fact that matters most. Explain why it matters.

## 02. Follow the deciding steps

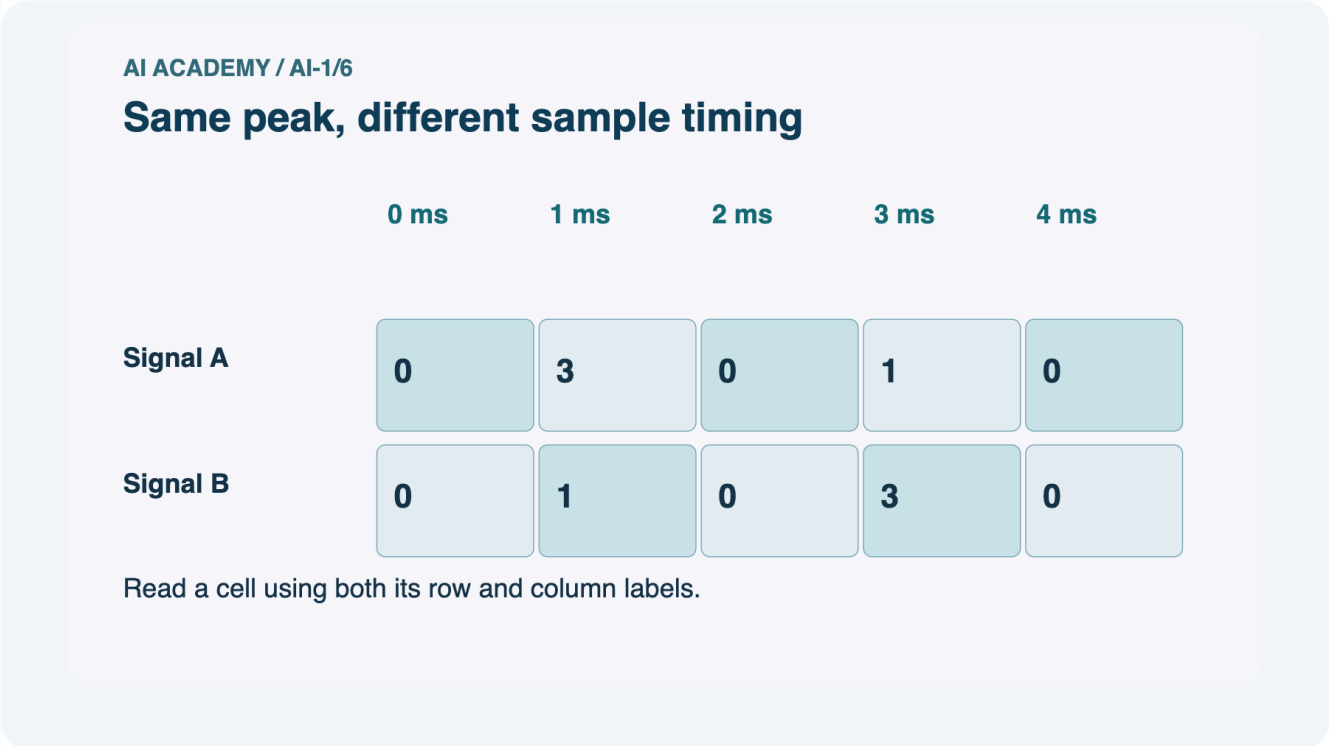



Follow the labelled picture with the matching explanation. Open the vector diagram

Timed sampling: Associate every measurement with its sampling time. → **Signal** A: Its largest **sample** 3 occurs at 1 ms. → **Signal** B: Its largest **sample** 3 occurs at 3 ms. → Maximum summary: Both maxima are 3, but the strongest recorded tap arrives at different times.

Point to each numbered stage and explain which recorded **input** it uses.

## 03. Same peak, different sample timing

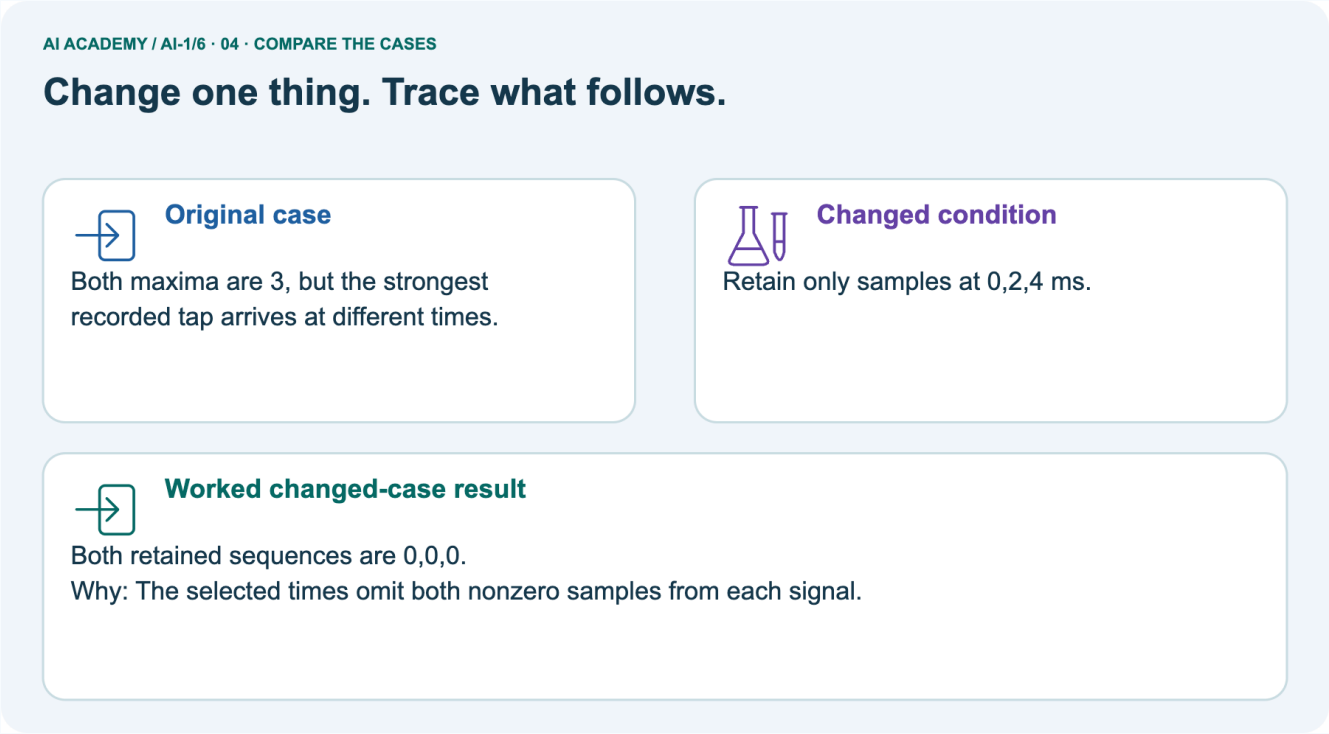



Follow the labelled picture with the matching explanation. Open the vector diagram

Same peak, different **sample** timing. **Signal** A: 0 ms = 0, 1 ms = 3, 2 ms = 0, 3 ms = 1, 4 ms = 0. **Signal** B: 0 ms = 0, 1 ms = 1, 2 ms = 0, 3 ms = 3, 4 ms = 0.

Read the labels and values. Explain the relationship the model shows.

## 04. Change one thing. Trace what follows.



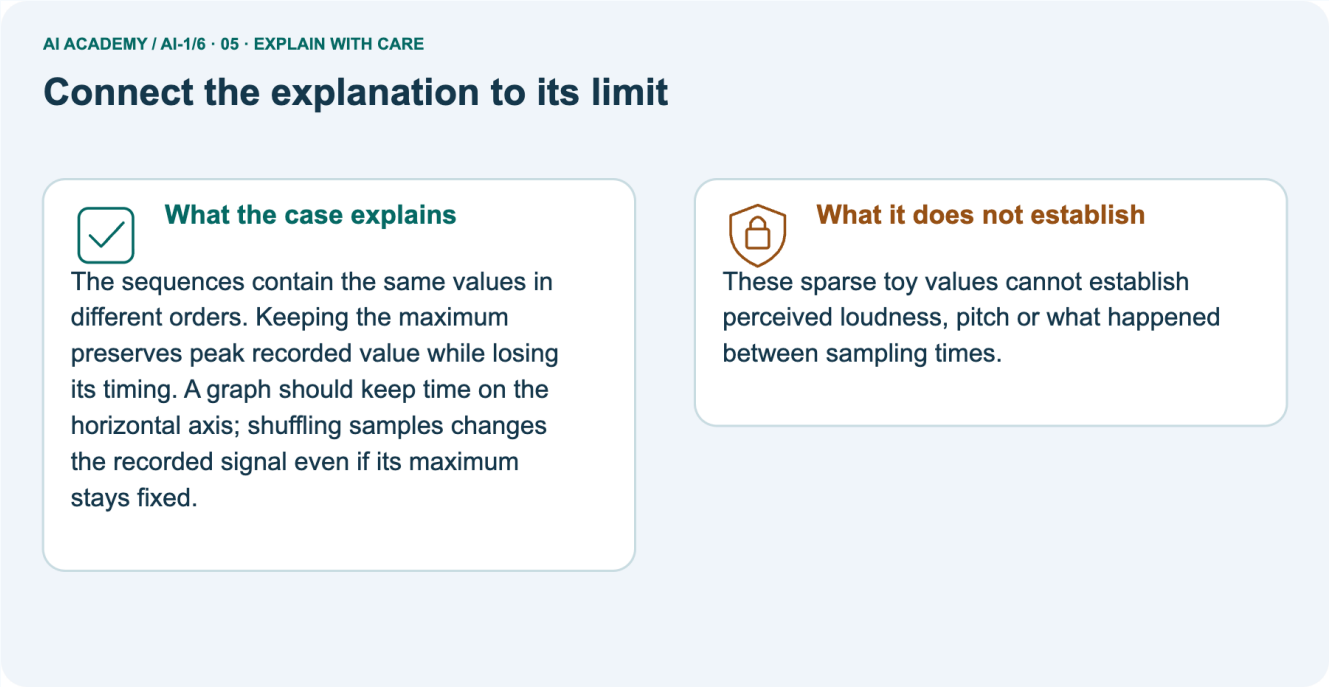

Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Both maxima are 3, but the strongest recorded tap arrives at different times. Changed condition: Retain only **sample**s at 0,2,4 ms. Expected result: Both retained sequences are 0,0,0.

Reason: The selected times omit both nonzero **sample**s from each **signal**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

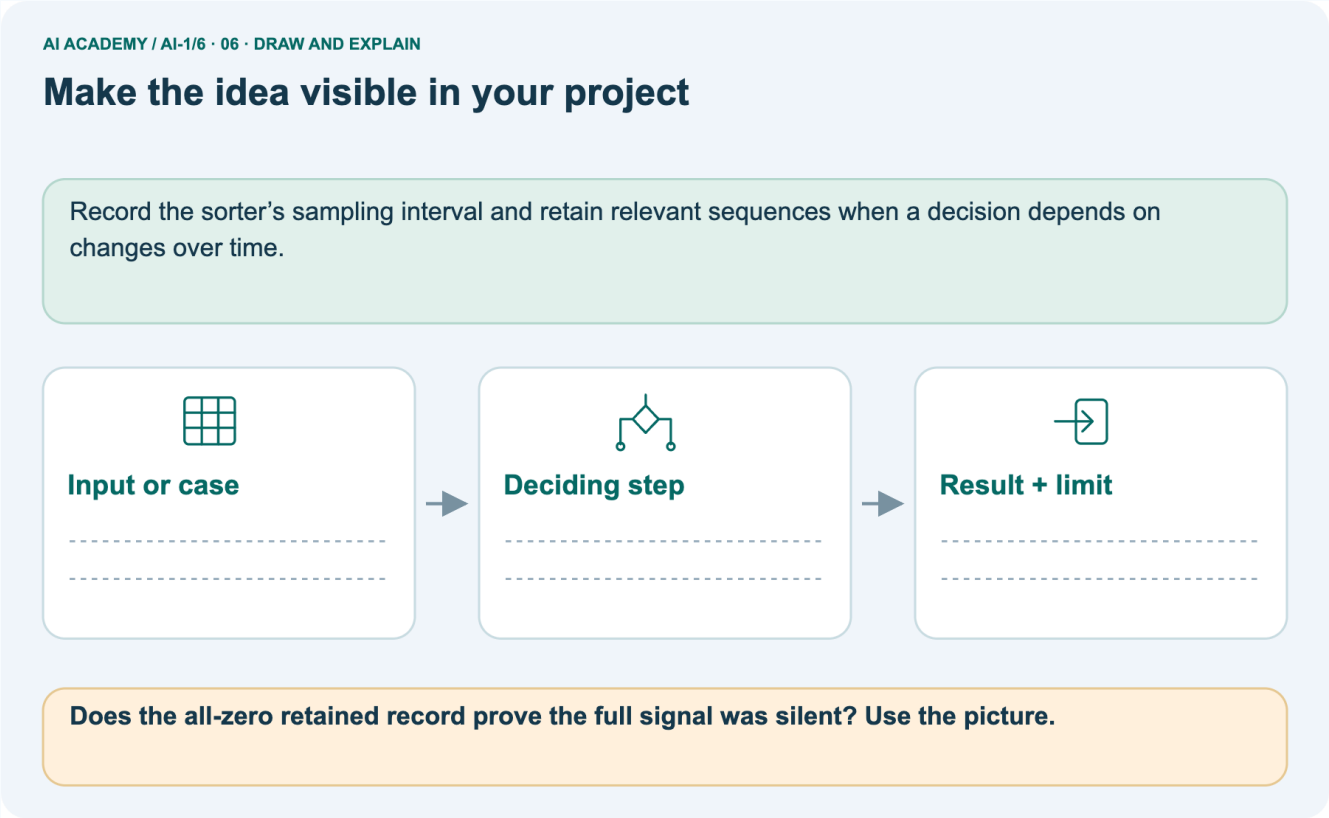
The sequences contain the same values in different orders. Keeping the maximum preserves peak recorded value while losing its timing.

A graph should keep time on the horizontal axis; shuffling **sample**s changes the recorded **signal** even if its maximum stays fixed.

Limit: These sparse toy values cannot establish perceived loudness, pitch or what happened between sampling times.

Explain the case, then point to a limit on the claim.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

Record the sorter's **sampling interval** and retain relevant sequences when a decision depends on changes over time. Draw your **input**, deciding step, result and limit.

Supported question: Does the all-zero retained record prove the full **signal** was silent? Use the picture.

Draw your project trace and justify the deciding step using **evidence**.

## ▼ Earlier wording, alternatives and picture transcripts

**01. Sound Becomes a Signal: the case**

**02. Follow the deciding steps**

**03. Same peak, different sample timing**

**04. Change one thing. Trace what follows.**

**05. Connect the explanation to its limit**

**06. Make the idea visible in your project**

## ▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.